RAS Prelims 2026 · Handwritten Class Notes

Geography of India

Chapter-by-chapter notes in coaching-lecture style — concept first, then the facts, with flow charts, mind maps, previous-year callouts and current-affairs links.

RPSC RAS & RTS Preliminary Examination · 6 December 2026 · 150 questions · 200 marks · 1/3 negative marking

IG01

Physiographic Divisions of India

About 1-2 questions a paper. Three documented repeats: the north-to-south sequence of Himalayan ranges (RAS 2024), the peninsular peaks Anamudi and Doddabetta (RAS 2024) and the Aravalli statement set (RAS 2023 and 2024).

Indian Geography gives you about 5 questions in the whole paper, and this chapter is the map half of it. Nothing here is analytical. It is a place matched to a state, a peak matched to a range, a pass matched to a road. Learn positions and heights, not descriptions.

Physiography of India

Extent and position
- 8 deg 4' N to 37 deg 6' N
- 68 deg 7' E to 97 deg 25' E
- 82 deg 30' E standard meridian
- Indira Point
- Tropic through 8 states

Himalaya, north to south
- Karakoram
- Ladakh
- Zaskar
- Himadri
- Himachal
- Shivalik

Peaks
- Kangchenjunga 8,586
- Nanda Devi 7,816
- Anamudi 2,695
- Doddabetta 2,637
- Guru Shikhar 1,722
- Arma Konda 1,680

Passes
- Zoji La
- Rohtang
- Shipki La
- Nathu La
- Se La
- Lipulekh

Northern Plains
- Bhabar
- Terai
- Bhangar
- Khadar
- Punjab doabs

Peninsular Plateau
- Aravalli
- Vindhya
- Satpura
- Malwa
- Chotanagpur
- Deccan Trap

Ghats and coasts
- Western Ghats continuous
- Eastern Ghats broken
- Palghat gap
- Konkan and Malabar
- Coromandel

Islands
- Barren Island
- Saddle Peak
- Ten Degree Channel
- Kavaratti
- Nine and Eight Degree Channels

Where India sits - the numbers RPSC actually asks

Start with position, because half the easy questions come from it. India stretches from 8 degrees 4 minutes North to 37 degrees 6 minutes North, and from 68 degrees 7 minutes East to 97 degrees 25 minutes East. The country is the 7th largest in the world with 32.87 lakh sq km, which is about 2.4 per cent of the world's land area. The east-west spread is so wide that the sun rises in Arunachal Pradesh roughly two hours before it rises in Gujarat. That is exactly why a single standard meridian was fixed.

- Standard Meridian of India = 82 degrees 30 minutes East; IST = GMT plus 5 hours 30 minutes.
- It passes through Mirzapur in Uttar Pradesh.
- It crosses five states - Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, Andhra Pradesh.
- Tropic of Cancer crosses eight states - Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram.
- Indira Point in Great Nicobar (6 deg 45' N) is the southernmost point of India.
- Kanniyakumari (8 deg 4' N) is the southernmost point of the mainland only - read the word 'mainland' before you answer.

- Seven land neighbours - Pakistan, Afghanistan, China, Nepal, Bhutan, Myanmar, Bangladesh.
- Two maritime neighbours - Sri Lanka and Maldives.
- Sri Lanka is separated from India by the Palk Strait and the Gulf of Mannar.

Borders and coastline - the figures worth memorising

Item	Figure
Total land frontier	15,106.7 km
Longest land border with	Bangladesh - 4,096.7 km
Shortest land border with	Afghanistan - 106 km
Coastline (re-assessed)	11,098.81 km
Earlier official coastline figure	7,516.6 km
State with longest coastline	Gujarat - 2,340.62 km

CURRENT LINK

Current link (as of September 2026): the coastline figure was re-assessed by the National Hydrographic Office to 11,098.81 km, against the long-quoted 7,516.6 km, because the newer survey follows the indented shoreline at a finer scale. Both numbers still float around in textbooks. If the option set contains 11,098.81 km, that is the newer official answer; if it does not, 7,516.6 km is what is being tested.

The Himalaya from north to south - the 2024 repeat

India has three geological divisions and they are of three different ages. The Peninsular block is the oldest - Archaean gneiss and granite, rigid and stable. The Himalaya is the youngest - folded sediments of the old Tethys sea, still rising. Between the two lies the Indo-Ganga plain, a trough filled with alluvium. Fix this three-way split first; everything else in the chapter hangs on it.

North to south across the Himalaya - learn this order

Karakoram (northernmost) → Ladakh → Zaskar → Great Himalaya / Himadri → Lesser Himalaya / Himachal → Shivalik (southernmost, youngest)

YAAD RAKHO

Trick to remember: Ka-La-Za-Hi-Hi-Shi. Karakoram, Ladakh, Zaskar, Himadri, Himachal, Shivalik. Say it downhill - you are walking from Tibet towards the plains, so the ranges get lower as you go.

ASKED IN RAS

Asked in RAS Pre 2024 - arrange Shivalik, Karakoram, Zaskar and Ladakh from north to south. Answer: Karakoram, Ladakh, Zaskar, Shivalik. The same paper also asked which range lies immediately north of the Great Himalaya. Answer: Zaskar.

The three main southern ranges

Range	Height	Marks of identity
Himadri (Great Himalaya)	about 6,000 m average	Carries all the great peaks; continuous
Himachal (Lesser Himalaya)	3,700 - 4,500 m	Pir Panjal, Dhauladhar, Mahabharat; hill stations
Shivalik	900 - 1,100 m	Southernmost and geologically youngest; unconsolidated

- Duns are longitudinal valleys between the Shivalik and the Lesser Himalaya - Dehra Dun, Kotli Dun, Patli Dun.
- Karewas are lacustrine (old lake-bed) deposits of the Kashmir valley, famous for saffron.
- Burrard's four regional divisions - Punjab Himalaya (Indus to Satluj), Kumaon Himalaya (Satluj to Kali).
- Then Nepal Himalaya (Kali to Teesta) and Assam Himalaya (Teesta to Dihang).
- Purvanchal hills - Patkai Bum, Naga, Manipur, Mizo (Lushai), Barail, Mikir, and the Garo-Khasi-Jaintia group.
- The Kaimur hills are NOT Purvanchal - they belong to the Vindhyan system of the Peninsular Plateau.

Peaks - and the trap RPSC set in 2024

Peaks are pure matching, so learn them as a height ladder with the state attached. Note the three careful phrasings the examiner uses. 'Highest peak of India' is Kangchenjunga. 'Highest peak lying entirely within Indian territory' is Nanda Devi, because Kangchenjunga sits on the border with Nepal. 'Highest peak in the Karakoram' is K2, which is higher than Kangchenjunga but lies in the Karakoram range.

Peaks by height, with the range and the state

Peak	Height	Where
K2 / Godwin Austen	8,611 m	Karakoram range
Kangchenjunga	8,586 m	Sikkim - highest of India, 3rd highest in the world
Nanda Devi	7,816 m	Chamoli, Uttarakhand - highest entirely within India
Kamet	7,756 m	Uttarakhand
Anamudi	2,695 m	Idukki-Ernakulam, Kerala - highest of Western Ghats and of peninsular India
Doddabetta	2,637 m	Nilgiri hills - highest peak of Tamil Nadu
Guru Shikhar	1,722 m	Mount Abu, Sirohi, Rajasthan - highest of the Aravalli
Arma Konda	1,680 m	Alluri Sitharama Raju district, Andhra Pradesh - highest of the Eastern Ghats
Dhupgarh	1,350 m	near Pachmarhi - highest of the Satpura
Saddle Peak	732 m	North Andaman - highest of the island group

ASKED IN RAS

Asked in RAS Pre 2024 - Anamudi, the highest peak of peninsular India, is in which state? Answer: Kerala. The same paper asked which hills Doddabetta belongs to. Answer: the Nilgiris.

TRAP

Do not write Anamudi as Tamil Nadu. Anamudi is in Kerala, where the Anaimalai, Cardamom and Palani hills meet. The highest peak in Tamil Nadu is Doddabetta. And the highest peak of the Eastern Ghats is Arma Konda, not Doddabetta - the Nilgiris are where the two Ghats join, they are not the Eastern Ghats. Peninsular ladder to fix it: Anamudi 2,695 - Doddabetta 2,637 - Guru Shikhar 1,722 - Arma Konda 1,680 - Dhupgarh 1,350; Kerala, Tamil Nadu, Rajasthan, Andhra, Madhya Pradesh.

Passes - a pass is a road, so learn what it connects

A pass question is never about height. It is about which state the pass is in and which two places it joins. Group them state by state and the matching becomes automatic.

Passes of the Himalaya

Pass	State / UT	Connects
Zoji La	Ladakh	Srinagar with Leh
Khardung La	Ladakh	Leh with the Nubra valley
Banihal	Jammu and Kashmir	Jammu with the Kashmir valley
Rohtang	Himachal Pradesh	Kullu with Lahaul and Spiti
Shipki La	Himachal Pradesh (Kinnaur)	India with Tibet; the Satluj enters India here
Bara Lacha La	Himachal Pradesh	On the Manali-Leh road
Kunzum	Himachal Pradesh	Lahaul with Spiti
Mana	Uttarakhand	Badrinath side with Tibet
Niti	Uttarakhand	India with Tibet
Lipulekh	Uttarakhand (Pithoragarh)	The Kailash-Mansarovar route
Nathu La	Sikkim	Gangtok with the Chumbi valley, Tibet

Pass	State / UT	Connects
Jelep La	Sikkim	Kalimpong side with Tibet
Se La and Bomdi La	Arunachal Pradesh	Tawang with the Assam plains
Diphu and Pangsau	Arunachal Pradesh	India with Myanmar

- Sort by state and the list stops being fourteen facts - two in Ladakh, four in Himachal, three in Uttarakhand, two in Sikkim, the rest in Arunachal.
- RAS has asked exactly this as a four-pair match: Shipki La - HP, Nathu La - Sikkim, Se La - Arunachal, Lipulekh - Uttarakhand.
- Zoji La is the one asked as a road link question - it joins Srinagar with Leh.

The Northern Plains - four belts, north to south

The Northern Plain is one long trough filled by the Indus, Ganga and Brahmaputra with their alluvium. Where the rivers leave the hills they drop coarse material, and further south they drop finer material. That single idea produces the four belts, and the belts are always asked in order.

The four belts of the Northern Plain, north to south

Bhabar - porous pebbles at the Shivalik foot; streams disappear underground → Terai - marshy and forested; the streams re-emerge →

Bhangar - old alluvium, above flood level, kankar nodules → Khadar - new alluvium of the flood plains, renewed almost every year

TRAP

Bhangar is the OLD alluvium that lies away from the flood plain and is not renewed. Khadar is the NEW alluvium of the flood plain and is enriched with fresh silt almost every year. RPSC swaps these two in a 'which statement is NOT correct' question. Bhangar has the kankar; Khadar has the fertility.

Punjab doabs, east to west

Doab	Lies between
Bist	Beas and Satluj
Bari	Beas and Ravi
Rechna	Ravi and Chenab
Chaj	Chenab and Jhelum
Sindh Sagar	Jhelum and Indus

- Doab names are built from the two rivers themselves - Rechna = Ravi plus Chenab, Chaj = Chenab plus Jhelum, Bari = Beas plus Ravi.
- The order east to west is Bist, Bari, Rechna, Chaj, Sindh Sagar.

The Peninsular Plateau and the two Ghats

The Peninsular Plateau is the oldest and most stable part of India - ancient crystalline rock, broadly triangular, tilted so that most of its rivers run east. Its edges are the Aravalli in the north-west, the Vindhya and Satpura across the middle, and the two Ghats down the sides. For RPSC the Aravalli is the highest-value item on this list, because it is Rajasthan's own range.

- Aravalli - the oldest fold mountain system of India, running north-east to south-west from Delhi to Gujarat.
- Highest Aravalli peak is Guru Shikhar (1,722 m), Mount Abu, Sirohi district.

- Malwa plateau lies between the Aravalli and the Vindhya.
- Dhupgarh (1,350 m) near Pachmarhi is the highest point of the Satpura.
- The Maikal range links the Satpura to the Chotanagpur plateau; Amarkantak is the source area of the Narmada, Son and Johilla.
- Chotanagpur plateau is drained by the Damodar; it is India's mineral heartland.
- Meghalaya plateau is cut off from the main block by the Malda gap, also called the Garo-Rajmahal gap.
- Deccan Trap is basaltic lava of Cretaceous-Eocene age; its weathering gives black regur or cotton soil.

ASKED IN RAS

Asked in RAS Pre 2023 and again in 2024 as a statement set on the Aravalli - it is one of the oldest fold mountain systems in India, it runs north-east to south-west, and its highest peak is Guru Shikhar in Sirohi district. All three statements are correct. Learn 'oldest fold mountain' as a fixed phrase; that is the wording the examiner reuses.

Western Ghats vs Eastern Ghats

Western Ghats (Sahyadri)	Eastern Ghats
Continuous wall	Discontinuous and broken
Crossed only through passes and ghats	Dissected by the east-flowing rivers
Higher - Anamudi 2,695 m	Lower - Arma Konda 1,680 m
Runs close and parallel to the west coast	Lies well inland of the east coast
Feeds short, swift, west-flowing streams	Cut by the Mahanadi, Godavari, Krishna, Kaveri

- The Eastern and Western Ghats meet at the Nilgiri hills.
- The Palghat (Palakkad) gap separates the Nilgiri from the Anaimalai hills.
- The Shencottah gap lies on the Kerala-Tamil Nadu border further south.

- Thal Ghat carries the Mumbai-Nashik route; Bhor Ghat carries the Mumbai-Pune route. Both are in Maharashtra.
- Hills to states: Cardamom - Kerala; Javadi - Tamil Nadu; Rajmahal - Jharkhand; Mukundara - Kota, Rajasthan.

TRAP

Bhor Ghat is in Maharashtra, not Karnataka. And note the Rajasthan hook in that hills list - Mukundara near Kota is the Rajasthan entry RPSC drops into an all-India matching set to see whether you are reading carefully.

Coastal plains and the islands

The two coasts behave in opposite ways, and that contrast is the whole question. The west coast is narrow because the Sahyadri comes right down to the sea, and it is a submergent coast - drowned, with lagoons and backwaters. The east coast is wide, emergent and deltaic, because the big peninsular rivers dump their load there.

- Western coastal plain, north to south: Kachchh-Kathiawar, Gujarat plain, Konkan, Kanara, Malabar.
- The Malabar backwaters are called kayals.
- Eastern coastal plain: Utkal (Odisha), Northern Circars / Andhra, Coromandel (Tamil Nadu).
- Chilika and Pulicat lagoons both lie along the eastern coastal plain.
- Barren Island is India's only active volcano; Narcondam is dormant. Both are in the Andamans.
- Lakshadweep is of coral origin, headquarters at Kavaratti.

Channels of the islands - a favourite match

Channel or passage	Separates
Ten Degree Channel	Little Andaman from Car Nicobar (Andaman from Nicobar group)
Duncan Passage	South Andaman from Little Andaman
Coco Channel	The Andamans from Myanmar's Coco islands
Nine Degree Channel	Minicoy from the main Lakshadweep group
Eight Degree Channel	Minicoy from the Maldives

YAAD RAKHO

Degrees go down as you go south: Ten Degree in the Andaman-Nicobar sea, then Nine Degree, then Eight Degree in the Lakshadweep sea. Nine separates Minicoy from India's own islands; Eight separates Minicoy from a foreign country, the Maldives. Higher number, nearer home.

REVISION RECAP

1. India: 8 deg 4' N to 37 deg 6' N, 68 deg 7' E to 97 deg 25' E; 32.87 lakh sq km; 7th largest; 2.4 per cent of world land.
2. Standard meridian 82 deg 30' E through Mirzapur and five states; IST = GMT + 5:30.
3. Indira Point is India's southernmost point; Kanniyakumari only of the mainland.
4. Tropic of Cancer crosses eight states; longest land border is with Bangladesh, shortest with Afghanistan.
5. North to south: Karakoram, Ladakh, Zaskar, Himadri, Himachal, Shivalik - asked in RAS 2024.
6. Himadri about 6,000 m; Himachal 3,700-4,500 m with Pir Panjal and Dhauladhar; Shivalik only 900-1,100 m.
7. Duns lie between Shivalik and Lesser Himalaya; Karewas are the lake deposits of Kashmir; Kaimur is Vindhyan, not Purvanchal.
8. Kangchenjunga highest in India; Nanda Devi highest entirely within India; K2 is in the Karakoram.
9. Anamudi (Kerala) highest in peninsular India; Doddabetta highest in Tamil Nadu; Arma Konda highest of the Eastern Ghats.
10. Guru Shikhar 1,722 m, Sirohi - highest of the Aravalli, the oldest fold mountain system of India.
11. Passes by state: Zoji La and Khardung La in Ladakh; Rohtang and Shipki La in HP; Lipulekh in Uttarakhand; Nathu La in Sikkim; Se La in Arunachal.

12. Plain belts north to south: Bhabar, Terai, Bhangar, Khadar. Bhangar is old and kankar-bearing, Khadar is new and renewed.
13. Doabs east to west: Bist, Bari, Rechna, Chaj, Sindh Sagar.
14. Western Ghats continuous and higher; Eastern Ghats broken and cut by rivers; the two meet at the Nilgiris.
15. Palghat gap separates Nilgiri from Anaimalai; Thal Ghat and Bhor Ghat are both in Maharashtra.
16. Deccan Trap basalt gives black regur soil; Meghalaya plateau is cut off by the Malda gap.
17. West coast narrow and submergent with kayals; east coast wide, emergent and deltaic with Chilika and Pulicat.
18. Barren Island is the only active volcano; Saddle Peak is the highest point of the Andamans; Kavaratti is the Lakshadweep headquarters.

IG02

Climate of India

About 1-2 questions a paper. Documented repeats: the retreating monsoon on the Tamil Nadu coast (RAS 2023 and 2024), rainfall variability highest in west Rajasthan and Ladakh (RAS 2024), and the Chennai-versus-Mumbai rainfall contrast (RAS 2024).

One chapter, one idea: the monsoon. Understand why the winds reverse and the rest of the chapter falls into place - onset dates, rain-shadows, variability, even the Tamil Nadu anomaly. RPSC does not ask you to explain the monsoon. It asks you for the fact that follows from the explanation, so learn the mechanism once and then drill the facts.

Climate of India

Mechanism
- Differential heating
- ITCZ shifts north
- Coriolis deflection
- Tibetan plateau heating

Jet streams
- Subtropical westerly
- Tropical easterly
- Somali (Findlater) jet

Four seasons (IMD)
- Winter Jan-Feb
- Pre-monsoon Mar-May
- SW monsoon Jun-Sep
- Post-monsoon Oct-Dec

Winter rain
- Western disturbances
- Mediterranean origin
- Mahawat
- Rabi wheat

Onset and withdrawal
- Kerala 1 June
- Whole country 8 July
- West Rajasthan 17 September
- Complete by 15 October

Retreating monsoon
- October heat
- Coromandel rain Oct-Dec
- Bay cyclones

Swing factors
- El Nino
- La Nina
- IOD
- SOI Tahiti-Darwin
- MONEX

Koppen types
- Amw west coast
- As Coromandel
- Aw peninsula
- BWhw west Rajasthan
- Cwg Ganga plain

What a monsoon is, and the machinery that drives it

A monsoon is a seasonal reversal of wind direction. In summer the land heats faster than the sea, a low pressure area forms over north-west India, and moist air is drawn in from the ocean. In winter the land cools faster, pressure rises over the interior, and dry air blows out towards the sea. That is the old, simple, thermal explanation, and it is still the first thing to say.

But heating alone does not explain why the rain arrives in a sudden burst in early June. The modern explanation adds three things in the upper air. One, the ITCZ - the equatorial low pressure trough - shifts north in summer and sits over the Ganga plain as the monsoon trough. Two, the south-east trade winds of the southern hemisphere cross the equator, get deflected to the right by the Coriolis force, and arrive over India as south-west winds. Three, the jet streams change position. Until the subtropical westerly jet moves out of the way, the monsoon cannot break.

The three jet streams and the trough - learn one line each

Wind system	What it does
Subtropical westerly jet	Lies south of the Himalaya in winter; shifts north of the Tibetan plateau in early June - a precondition for the burst
Tropical easterly jet	Develops over peninsular India in summer, above the intensely heated Tibetan plateau
Somali (Findlater) jet	Low-level cross-equatorial jet off the East African coast; strengthens the Arabian Sea branch
Monsoon trough	The northward-shifted ITCZ lying over the Ganga plain in July

Why the monsoon bursts in early June

Land heats, low pressure deepens over north-west India →

Subtropical westerly jet withdraws north of the Tibetan plateau →

Tropical easterly jet establishes over the peninsula →

Somali jet strengthens the Arabian Sea branch →

South-east trades cross the equator and turn south-west →

Burst of the monsoon over Kerala, about 1 June

YAAD RAKHO

Two jets, two directions, two seasons. Westerly jet = winter feature, brings western disturbances. Easterly jet = summer feature, sits over the peninsula. If a question puts western disturbances with the easterly jet, it is wrong.

Winter, and the western disturbances that feed the rabi crop

Winter in India should be bone dry, because the winds blow from land to sea. Yet Punjab, Haryana, western UP and northern Rajasthan get a little rain in December to February, and that small rain decides the wheat crop. The reason is the western disturbance - an extra-tropical cyclone that forms over the Mediterranean region and is dragged across Iran, Afghanistan and Pakistan into India by the subtropical westerly jet stream.

- *Western disturbances are extra-tropical cyclones of Mediterranean origin.*
- *They are steered into India by the subtropical westerly jet, not by any monsoon wind.*
- *The winter rain they bring is locally called mahawat.*
- *Mahawat is small in amount but decisive for the rabi wheat crop.*

- They also cause snowfall in the western Himalaya and occasional hailstorms in Rajasthan.
- Their track weakens as they move east, so eastern India gets almost nothing from them.

IMD's four seasons

Season	Months
Winter	January and February
Pre-monsoon / hot weather	March to May
South-west monsoon	June to September
Post-monsoon / retreating monsoon	October to December

TRAP

Western disturbances originate over the Mediterranean, not the Arabian Sea and not the Bay of Bengal. And they are carried by the subtropical westerly jet, not the tropical easterly jet. RPSC has used both of these as the wrong option in a 'NOT correct' question.

- Say the chain as one phrase - Mediterranean, westerly jet, mahawat, wheat.
- Four words, one chain, and three separate RAS questions answered.

Hot weather season and the local winds

March to May the interior bakes. Temperature rises, pressure falls, and before the monsoon arrives the country gets a scatter of local winds and thunderstorms with regional names. These names are pure matching material - learn wind to region, nothing else.

Local winds and pre-monsoon showers

Name	Region
Loo	Northern plains, May-June - hot dry westerly
Kal Baisakhi	West Bengal - violent pre-monsoon nor'wester
Bardoli Chheerha	Assam - the nor'wester of the Brahmaputra valley
Mango showers	Kerala and coastal Karnataka
Blossom showers	Coffee areas of Karnataka

TRAP

Bardoli Chheerha belongs to Assam, not Punjab. That exact wrong pair has been used in a 'NOT correctly matched' question. Kal Baisakhi is Bengal, Bardoli Chheerha is Assam - two nor'westers, two different states.

- Mango showers help the mango crop ripen in Kerala and coastal Karnataka.
- Blossom showers help coffee flowers to set in Karnataka.
- The Loo is a heat wind, not a rain-bearing wind.

Onset, advance and withdrawal - the calendar

The monsoon does not arrive everywhere at once. It bursts on the Kerala coast, splits into two branches around the peninsula, and sweeps north-west over about five weeks. It then retreats in the reverse direction - out of the north-west first, out of the far south last. Get the direction of advance and retreat right and the dates follow.

Onset and advance of the south-west monsoon

Kerala coast - normal onset 1 June → Splits into the Arabian Sea branch and the Bay of Bengal branch → Arabian Sea branch strikes the Western Ghats; Bay branch is turned west by the Himalaya → The two branches converge over the north-western plains → Whole country covered by about 8 July

Withdrawal - the reverse order

Begins from west Rajasthan - IMD revised normal date 17 September → Retreats from the north-western plains through late September → Clears the Ganga plain and central India in October → Withdrawal from the whole country complete by about 15 October → Winds now blow off the land towards the Bay - the north-east monsoon

- IMD revised the normal start of withdrawal from west Rajasthan from 1 September to 17 September.
- The Arabian Sea branch gives heavy rain on the windward western slopes of the Western Ghats and leaves a rain-shadow on the Deccan plateau.
- Mawsynram in the Khasi hills of Meghalaya has the world's highest average annual rainfall; funnel-shaped valleys trap the Bay branch.
- A 'break' in the monsoon is a dry spell within the rainy season, caused by a shift in the position of the monsoon trough.
- West Rajasthan (Jaisalmer, Barmer) and Ladakh receive less than 20 cm of rain; Leh is a cold desert.

- Advance is south-east to north-west; withdrawal is north-west to south-east.
- The monsoon leaves first from where it arrived last - that one line answers the direction question in either form.

The retreating monsoon and the Tamil Nadu coast - the biggest repeat here

This is the part of the chapter RPSC comes back to. Tamil Nadu is the odd one out in Indian climate. It stays dry when the rest of India is drenched, and it gets its rain when the rest of India is drying out. Two reasons, and you must be able to give both. First, the Coromandel coast lies in the rain-shadow of the Western Ghats, so the Arabian Sea branch has already lost its moisture by the time it crosses. Second, the coast runs parallel to the Bay of Bengal branch instead of standing across it, so that branch does not strike it either.

Then in October the winds reverse. The retreating north-east monsoon blows from land to sea, but on its way to Tamil Nadu it crosses the Bay of Bengal, picks up moisture, and delivers it on the Coromandel coast in October to December. That is why Chennai's rainiest months are October and November.

ASKED IN RAS

Asked in RAS Pre 2023 and again in 2024 - the Tamil Nadu coast receives the bulk of its annual rainfall in October to December from which system? Answer: the retreating or north-east monsoon.

ASKED IN RAS

Asked in RAS Pre 2023, also framed as assertion-reason - why is the Coromandel coast dry during the south-west monsoon? Answer: it lies in the rain-shadow of the Western Ghats and runs parallel to the Bay of Bengal branch.

Mumbai vs Chennai - the contrast RPSC asked in 2024

Mumbai (west coast)	Chennai (Coromandel coast)
Over 2,000 mm a year	About 1,200 to 1,400 mm a year
Rain concentrated June to September	Maximum rain in October and November
Fed by the Arabian Sea branch of the SW monsoon	Fed mainly by the retreating north-east monsoon
Windward side of the Western Ghats	Rain-shadow side of the Western Ghats
Low rainfall variability, below 25 per cent	Higher variability, dependent on a short season

- The retreating monsoon season brings the oppressive 'October heat' - high temperature with high humidity and clear skies.
- Tropical cyclones are frequent over the Bay of Bengal in this season.
- The withdrawal of the monsoon begins from north-western India, not from the south.

Rainfall distribution, variability and Koppen's regions

One rule explains the whole variability map: variability is inversely related to amount. Where rainfall is heavy it is also dependable; where rainfall is scanty it is also erratic. That is why west Rajasthan, which gets least, also suffers the wildest swings - and why droughts are a Rajasthan problem and not a Kerala problem.

- Variability above 50 per cent - west Rajasthan, Ladakh, the interior Deccan.
- Variability below 25 per cent - the west coast, the north-east, the Himalaya.
- Rainfall variability is inversely related to rainfall amount.
- Mawsynram highest, west Rajasthan and Ladakh lowest - the two extremes of the same map.

ASKED IN RAS

Asked in RAS Pre 2024 - the coefficient of variability of annual rainfall in India is highest in which region? Answer: western Rajasthan and Ladakh. Note the Rajasthan angle; RPSC will nearly always pick the option that carries the state's name when the fact allows it.

Koppen's climatic types in India

Code	Type and region
Amw	Monsoon with short dry season - the west (Malabar) coast
As	Monsoon with dry summer - the Coromandel coast
Aw	Tropical savanna - most of the peninsula
BShw	Semi-arid steppe
BWhw	Hot desert - west Rajasthan
Cwg	Monsoon with dry winter - the Ganga plain
Dfc	Cold humid winter with short summer - Arunachal Pradesh
E	Polar - the high Himalaya

TRAP

Arunachal Pradesh is Dfc, not Aw. Aw is the tropical savanna of the peninsula. Also keep As (Coromandel, dry summer) apart from Amw (west coast, short dry season) - the two coasts, two codes, and RPSC has matched them both.

What swings the monsoon, and the hazards it leaves behind

The monsoon's strength changes from year to year, and the drivers sit in the oceans. El Nino is an abnormal warming of the eastern Pacific, and it usually goes with a weak Indian monsoon. La Nina is the opposite phase and usually goes with a strong one. The Indian Ocean Dipole works closer to home: when the western Indian Ocean is warmer than the east, the dipole is positive and the monsoon does well - strongly enough at times to cancel out an El Nino.

- El Nino - warm eastern Pacific - weak or deficient Indian monsoon.
- La Nina - above-normal monsoon rainfall over India.
- ENSO is tracked by the Southern Oscillation Index - the pressure difference between Tahiti and Darwin.
- Positive Indian Ocean Dipole - warmer western Indian Ocean - favours a good monsoon and can offset an El Nino.
- MONEX, the Monsoon Experiment, ran under the Global Weather Experiment.
- Winter MONEX December 1978; Summer MONEX May to July 1979, over the Arabian Sea and Bay of Bengal.

Cyclones, droughts, floods - the hazard facts

Item	Fact
Bay vs Arabian Sea cyclones	Bay of Bengal cyclones are about four times as frequent
Most cyclone-prone coast	The Odisha-Andhra Pradesh coast
Cyclone peaks	Just before and just after the south-west monsoon
Naming authority	IMD as the RSMC New Delhi, for thirteen WMO/ESCAP member countries
Moderate drought	Rainfall deficiency of 26 to 50 per cent of normal
Severe drought	Rainfall deficiency of more than 50 per cent of normal

Item	Fact
Most flood-prone belts	Brahmaputra valley, the Kosi belt of north Bihar, the Damodar basin

TRAP

A deficiency of 26 to 50 per cent is a moderate drought - not 10 to 25 per cent. RPSC has planted the 10-25 figure as the wrong statement in a statement-based question. Learn the two thresholds as 26 and 50.

CURRENT LINK

Current link (as of September 2026): IMD, founded in 1875 and headquartered at New Delhi, completed 150 years on 15 January 2025, and Mission Mausam was launched around that anniversary to upgrade India's weather observation and modelling. Expect the founding year 1875, the 150-year milestone and Mission Mausam to appear as a current-affairs one-liner.

REVISION RECAP

1. Monsoon = seasonal reversal of winds; thermal low over north-west India plus the northward shift of the ITCZ.
2. South-east trades cross the equator and are deflected right by the Coriolis force to become south-west monsoon winds.
3. Subtropical westerly jet must withdraw north of the Tibetan plateau before the monsoon can burst.
4. Tropical easterly jet is a summer feature over the peninsula; the Somali (Findlater) jet strengthens the Arabian Sea branch.
5. Western disturbances - extra-tropical cyclones of Mediterranean origin, carried by the westerly jet; their rain is mahawat and it saves the rabi wheat.
6. IMD seasons: winter Jan-Feb, pre-monsoon Mar-May, south-west monsoon Jun-Sep, post-monsoon Oct-Dec.
7. Onset over Kerala 1 June; whole country covered by about 8 July.
8. Withdrawal begins from west Rajasthan on the revised normal date of 17 September; complete by about 15 October.

9. Arabian Sea branch soaks the windward Western Ghats and leaves the Deccan in rain-shadow; Mawsynram records the world's highest average rainfall.
10. Tamil Nadu is dry in the south-west monsoon - rain-shadow plus a coast parallel to the Bay branch.
11. The retreating north-east monsoon waters the Coromandel coast in October-December; October heat marks the season.
12. Mumbai over 2,000 mm in June-September; Chennai 1,200-1,400 mm peaking in October-November.
13. Variability is inversely related to amount - above 50 per cent in west Rajasthan and Ladakh, below 25 per cent on the west coast.
14. El Nino weakens the monsoon, La Nina strengthens it, a positive IOD helps and can offset El Nino; SOI = Tahiti minus Darwin.
15. MONEX 1978-79 studied the monsoon under the Global Weather Experiment.
16. Koppen: Amw west coast, As Coromandel, Aw peninsula, BWhw west Rajasthan, Cwg Ganga plain, Dfc Arunachal, E high Himalaya.
17. Bay cyclones about four times as frequent as Arabian Sea ones; Odisha-Andhra coast most vulnerable; IMD names them as RSMC New Delhi.
18. Moderate drought = 26-50 per cent deficiency; severe = above 50 per cent; IMD founded 1875.

IG03

Rivers, Lakes and Drainage of India

About 1-2 questions a paper. The locked repeat is tributary-to-main-river matching (RAS 2013, 2021, 2024 - Rapti, Teesta, Hiran, Purna). Ramsar sites and the Indus Waters Treaty are the other two live zones.

Rivers are the single most reliable scoring block in Indian Geography, because the questions are always the same shape - a tributary matched to its main river, a lake matched to its state, a waterfall matched to its river. RPSC has asked the tributary match in 2013, 2021 and 2024. Build the tables once and this chapter never troubles you again.

Drainage of India

Himalayan systems
- Indus
- Ganga
- Brahmaputra

Indus
- Bokhar Chu glacier
- Zaskar, Shyok, Nubra
- Treaty 1960
- Eastern rivers to India

Ganga
- Devprayag
- Panch Prayag
- Left bank - Ghaghara, Gandak, Kosi
- Right bank - Yamuna, Son

Brahmaputra
- Tsangpo
- Siang/Dihang
- Jamuna
- Majuli

West-flowing
- Narmada
- Tapi
- Mahi
- Sabarmati
- Luni

East-flowing
- Mahanadi
- Godavari
- Krishna
- Kaveri

Lakes
- Wular
- Vembanad
- Chilika
- Loktak
- Sambhar
- Lonar

Policy
- Ramsar 1971
- NPP 1980
- NWDA 1982
- Ken-Betwa

Two families of rivers - and why they behave differently

Every Indian river belongs to one of two families, and the difference is not decorative. Himalayan rivers rise in snow, so they flow all year. They are also older than the mountains in places - they kept cutting downward while the Himalaya rose around them, which is why they slice deep gorges through the ranges. That is called an antecedent course. Peninsular rivers rise in rainfall alone, so they shrink to a trickle in the dry season, and they have been flowing so long over stable rock that their valleys are broad, shallow and nearly graded.

Himalayan vs Peninsular rivers

Himalayan rivers	Peninsular rivers
Perennial - fed by snowmelt and rain both	Seasonal - rain-fed only
Antecedent in places; cut deep gorges	Flow in broad, shallow, graded valleys
Long courses with large basins and heavy silt	Shorter courses, smaller basins, less silt
Build large deltas - the Ganga-Brahmaputra delta is the world's largest	Mostly deltas in the east; Narmada and Tapi make estuaries
Meander freely and shift course over the plains	Fixed courses on hard rock
Navigable in the plains; support canal irrigation	Mostly non-navigable; suited to dam sites

- The Ganga basin is India's largest river basin - about a quarter of the country's area.
- The Godavari basin is the largest peninsular basin; the Godavari is the longest peninsular river.
- Length order to remember: Ganga 2,525 km, Godavari 1,465 km, Krishna 1,400 km, Narmada 1,312 km.
- Anchor any descending-order question on that four-river ladder.

The Indus system and the Indus Waters Treaty

The Indus rises near the Bokhar Chu glacier in the Kailash range of Tibet and enters India in Ladakh. It is the only major Indian river that flows mostly outside India today, which is exactly why its waters had to be shared by treaty. The treaty split the system geographically - three eastern rivers to India, three western rivers to Pakistan.

- Himalayan tributaries of the Indus - Zaskar, Shyok, Nubra, Hunza, Gilgit.
- Jhelum rises from a spring at Verinag in Anantnag and flows through Wular lake.
- Chenab is formed by the union of the Chandra and the Bhaga at Tandi in Himachal Pradesh.
- Indus Waters Treaty signed at Karachi on 19 September 1960 by Jawaharlal Nehru and Ayub Khan.
- The World Bank is a signatory to the treaty; the Permanent Indus Commission implements it.
- Rajasthan link - the Indira Gandhi Canal draws on the Satluj-Beas share that this treaty gave India.

Who gets which rivers

Group	Rivers	Allotted to
Eastern rivers	Ravi, Beas, Satluj	India
Western rivers	Indus, Jhelum, Chenab	Pakistan

CURRENT LINK

Current link (as of September 2026): India placed the Indus Waters Treaty in abeyance in April 2025 following the Pahalgam terror attack, and has since reaffirmed that position; the dispute is before the Permanent Court of Arbitration. For prelims, the safe core remains the treaty facts - 1960, Karachi, World Bank, eastern rivers to India - with the abeyance as the current-affairs tail.

The Ganga - headwaters, prayags, tributaries

The Ganga is not born as the Ganga. The Bhagirathi comes from Gaumukh at the snout of the Gangotri glacier, the Alaknanda comes from Satopanth, and the two meet at Devprayag - only below that point is the river called the Ganga. It enters the plains at Haridwar. Along the Alaknanda are five confluences, the Panch Prayag, and RPSC has asked them as an upstream-to-downstream ordering question.

Panch Prayag on the Alaknanda, upstream to downstream

Vishnuprayag - Dhauliganga joins → Nandprayag - Nandakini joins → Karnaprayag - Pindar joins → Rudraprayag - Mandakini joins → Devprayag - Bhagirathi joins; the Ganga begins

Ganga tributaries by bank

Bank	Tributaries
Left bank	Ramganga, Gomti, Ghaghara, Gandak, Burhi Gandak, Kosi, Mahananda
Right bank	Yamuna, Son, Punpun

- Yamuna rises from the Yamunotri glacier on Bandarpunch; its longest tributary is the Tons.
- Peninsular tributaries of the Yamuna - Chambal, Sind, Betwa, Ken. The Chambal is Rajasthan's river.
- Ghaghara is the Karnali in Nepal; the Rapti and the Sarda (Kali) are its tributaries.
- Kosi, the 'Sorrow of Bihar', is formed by the Sun Kosi, Arun and Tamur, jointly the Sapta Kosi.
- Damodar, the 'Sorrow of Bengal', drains the Chotanagpur plateau and joins the Hooghly.
- Damodar Valley Corporation (1948) was India's first multipurpose river valley project.

TRAP

The Son is a RIGHT-bank tributary of the Ganga, not a left-bank one - it rises at Amarkantak and comes up from the south. RPSC has used exactly this in a 'which is NOT a left-bank tributary' question. Only the Yamuna, Son and Punpun join from the right - the three that come up from the south. Everything else on the standard list - Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda - comes down from the Himalaya on the left.

The tributary table RPSC keeps coming back to

This is the highest-value table in the chapter. The same four pairs have been recycled across papers, and the format is always the same - a small tributary in List I, four main rivers in List II. Learn the table, not the reasoning.

Tributary to main river - the documented repeats

Tributary	Joins
Rapti	Ghaghara
Teesta	Brahmaputra
Hiran	Narmada
Purna (Marathwada)	Godavari
Tons	Yamuna
Chambal, Sind, Betwa, Ken	Yamuna
Sabari, Indravati, Manjra, Pranhita	Godavari
Bhima, Koyna, Tungabhadra, Musi	Krishna
Kabini, Hemavati, Bhavani, Amaravati	Kaveri
Tawa (largest), Burhner, Banjar, Orsang	Narmada
Girna, Panjhra, Waghur, Bori	Tapi
Subansiri, Kameng, Manas, Dhansiri, Kopili	Brahmaputra

ASKED IN RAS

Asked in RAS Pre 2021 and 2024 as a four-pair match - Rapti to Ghaghara, Teesta to Brahmaputra, Hiran to Narmada, Purna to Godavari. The individual pairs have also been asked as single one-liners in 2024. If you memorise one table in this chapter, make it this one.

TRAP

There are two Purnas. One is a tributary of the Tapi. The other, which rises in the Ajanta hills of Chhatrapati Sambhajinagar (Aurangabad) district, joins the Godavari - and that is the one the exam has asked. Read the descriptor in the question before you answer. Second pair to guard: the Sabari belongs to the Godavari, not the Krishna - that has been the wrong option in a NOT-correctly-matched question.

The Brahmaputra

The Brahmaputra changes its name four times on its way to the sea, and the name sequence is what gets asked. It also carries an enormous silt load, which is why it braids across the Assam valley and why it built Majuli, the largest river island in India.

The Brahmaputra's names, source to sea

Tsangpo in Tibet → Siang or Dihang on entering Arunachal Pradesh → Brahmaputra in Assam, after the Dibang and Lohit join → Jamuna in Bangladesh → Meghna after merging with the Padma (the Ganga)

- Tributaries - Subansiri (antecedent), Kameng, Manas, Sankosh, Teesta, Dhansiri, Kopili.
- Majuli in Assam is the largest river island of India.
- The Brahmaputra valley of Assam is among India's most flood-prone regions.

ASKED IN RAS

Asked in RAS Pre 2021 - the Teesta is a tributary of which river? Answer: the Brahmaputra. Do not let the Sikkim location pull you towards the Ganga; the Teesta joins the Brahmaputra in Bangladesh.

TRAP

In Bangladesh the Brahmaputra is the Jamuna, and the Ganga is the Padma. Below their junction the combined stream is the Meghna. RPSC has set 'the Brahmaputra is called the Padma in Bangladesh' as the false statement.

Peninsular rivers - west-flowing and east-flowing

The peninsula is tilted eastwards, so most of its rivers run to the Bay of Bengal. Only a handful run west, and those are the exceptions worth learning first. The Narmada and the Tapi are special because they do not flow in ordinary valleys at all - they occupy rift valleys, downfaulted troughs, and a rift valley river carries its silt straight out to sea instead of spreading it. That is why they make estuaries in the Gulf of Khambhat and not deltas.

- Narmada (1,312 km) rises at Amarkantak and flows west between the Vindhya and the Satpura.
- Dhuandhar falls and the Marble Rocks gorge at Bhedaghat are near Jabalpur on the Narmada.
- Tapi (724 km) rises at Multai in Betul district of Madhya Pradesh, also in a rift valley.
- Mahi is the only Indian river that crosses the Tropic of Cancer twice.
- Sabarmati and Luni both rise in the Aravalli.
- Luni turns saline below Balotra and ends in the Rann of Kachchh - Rajasthan's own west-flowing river.
- Godavari (1,465 km), the 'Dakshin Ganga', rises at Trimbakeshwar near Nashik.
- Krishna (1,400 km) rises near Mahabaleshwar; Kaveri (800 km) at Talakaveri in the Brahmagiri hills of Kodagu, Karnataka.

- Mahanadi rises near Sihawa in Dhamtari district of Chhattisgarh and is dammed at Hirakud.
- Odisha's other rivers - Subarnarekha, Brahmani, Baitarani.

River to place of origin - a standard match

River	Rises at
Godavari	Trimbakeshwar, Nashik
Krishna	Near Mahabaleshwar
Kaveri	Talakaveri, Brahmagiri hills, Kodagu
Mahanadi	Sihawa, Dhamtari, Chhattisgarh
Tapi	Multai, Betul, Madhya Pradesh
Narmada and Son	Amarkantak

Waterfalls - river and state

Waterfall	River	State
Kunchikal	Varahi	Karnataka
Jog (Gersoppa)	Sharavathi	Karnataka
Dudhsagar	Mandovi	Goa
Chitrakote	Indravati	Chhattisgarh
Hundru	Subarnarekha	Jharkhand
Athirappilly	Chalakudy	Kerala
Shivanasamudram	Kaveri	Karnataka
Nohkalikai	-	Meghalaya

TRAP

Two standard traps here. One - the Narmada and the Tapi do NOT build deltas; they form estuaries because they flow through rift valleys. Two - Chitrakote falls are in Chhattisgarh on the Indravati, not in Odisha. Chitrakote is the widest fall in India.

Lakes, Ramsar sites and river interlinking

Lake questions are almost entirely superlative-plus-state. Learn which lake holds which record, and where it is. Note how many of these superlatives Rajasthan owns - Sambhar is India's largest inland saline lake and Keoladeo was one of the country's first two Ramsar sites, so RPSC has a permanent reason to keep coming back to this section.

Lakes - the record and the state

Lake	Record	State / UT
Wular	Largest freshwater lake of India; lies on the Jhelum	Bandipora, J&K
Vembanad	Longest lake in India	Kerala
Chilika	Largest brackish-water lagoon; holds Nalabana Bird Sanctuary	Odisha
Pulicat	Second largest lagoon; Sriharikota lies behind it	Andhra Pradesh - Tamil Nadu
Loktak	Floating phumdi mats; Keibul Lamjao, the only floating national park	Manipur
Kolleru	Lies between the Krishna and Godavari deltas	Andhra Pradesh
Sambhar	Largest inland saline lake of India; a Ramsar site	Rajasthan
Lonar	Meteorite-impact crater lake in Deccan basalt	Buldhana, Maharashtra
Roopkund	The 'Skeleton Lake', about 5,029 m, below Trisul	Chamoli, Uttarakhand
Kanwar (Kabartal)	Large freshwater oxbow lake	Begusarai, Bihar

YAAD RAKHO

Four superlatives, four different words - largest FRESHWATER is Wular, LONGEST is Vembanad, largest BRACKISH lagoon is Chilika, largest inland SALINE is Sambhar. The examiner changes only the adjective. Read the adjective.

- Ramsar Convention signed in 1971 at Ramsar, Iran; India acceded in 1982.
- Chilika (Odisha) and Keoladeo Ghana (Rajasthan) were designated in 1981 as India's first Ramsar sites.
- Sundarban Wetland is India's largest Ramsar site; Renuka in Himachal Pradesh is among the smallest.
- Tamil Nadu has the largest number of Ramsar sites among Indian states.
- Keoladeo and Sambhar are Rajasthan's Ramsar sites - both have been asked directly by RPSC.

ASKED IN RAS

Asked in RAS Pre 2016 and again in 2023 - which two wetlands were designated India's first Ramsar sites in 1981? Answer: Chilika and Keoladeo Ghana. Note that India acceded to the Convention only in 1982, a year after these two were designated - that gap is itself examinable.

CURRENT LINK

Current link (as of September 2026): India's Ramsar tally has risen to 101 sites, the most recent being Glaw Lake in Arunachal Pradesh, designated on 3 August 2026. The count moves, so carry the as-of date; the fixed facts - 1971 Ramsar Iran, India acceded 1982, first sites Chilika and Keoladeo in 1981 - are what a prelims question can safely rest on.

- National Perspective Plan (1980) proposes 30 links - 16 peninsular and 14 Himalayan.

- National Water Development Agency (NWDA) was set up in 1982.
- Ken-Betwa is the first interlinking project taken up; it involves the Daudhan dam, close to the Panna Tiger Reserve.
- Inter-State River Water Disputes Act, 1956 governs the tribunals.
- Tribunals so far - Kaveri, Krishna, Godavari, Narmada (award 1979), Ravi-Beas (Eradi), Mahadayi, Vansadhara.

REVISION RECAP

1. Himalayan rivers are perennial, antecedent and gorge-cutting; peninsular rivers are seasonal with broad graded valleys.
2. Ganga basin is India's largest; Godavari basin the largest peninsular one and the Godavari the longest peninsular river.
3. Indus rises near Bokhar Chu in the Kailash range; Jhelum at Verinag; Chenab from Chandra plus Bhaga at Tandi.
4. Indus Waters Treaty 1960 at Karachi, Nehru and Ayub Khan, World Bank a signatory, Permanent Indus Commission implements it.
5. Eastern rivers Ravi-Beas-Satluj to India; western Indus-Jhelum-Chenab to Pakistan; India held the treaty in abeyance from April 2025.
6. Ganga formed at Devprayag from the Bhagirathi and Alaknanda; enters the plains at Haridwar.
7. Panch Prayag order: Vishnuprayag, Nandprayag, Karnaprayag, Rudraprayag, Devprayag.
8. Right-bank tributaries of the Ganga are only Yamuna, Son and Punpun; the rest are left-bank.
9. Repeat table: Rapti to Ghaghara, Teesta to Brahmaputra, Hiran to Narmada, Purna to Godavari.
10. Brahmaputra names: Tsangpo, Siang/Dihang, Brahmaputra, Jamuna, Meghna; Majuli is India's largest river island.
11. Narmada from Amarkantak and Tapi from Multai flow in rift valleys and make estuaries, not deltas.
12. Mahi crosses the Tropic of Cancer twice; Luni rises in the Aravalli and turns saline below Balotra.
13. Kosi is the Sorrow of Bihar, Damodar the Sorrow of Bengal; DVC 1948 was India's first multipurpose project.
14. Jog on the Sharavathi, Kunchikal on the Varahi, Dudhsagar on the Mandovi, Chitrakote on the Indravati, Hundru on the Subarnarekha.

15. Wular largest freshwater, Vembanad longest, Chilika largest brackish lagoon, Sambhar largest inland saline.
16. Loktak carries phumdis and Keibul Lamjao; Lonar is a meteorite crater lake; Roopkund the Skeleton Lake in Chamoli.
17. Ramsar 1971 Iran, India acceded 1982, first sites Chilika and Keoladeo in 1981, Sundarban largest, Tamil Nadu has the most.
18. NPP 1980 with 30 links, NWDA 1982, Ken-Betwa the first project with the Daudhan dam.

IG04

Irrigation and Multipurpose Projects of India

About 1 question a paper from the Indian Geography slot, but the same material comes back a second time through the Rajasthan Geography slot - IGNP and the Chambal projects are asked almost every year there. Atal Bhujal Yojana alone was asked in 2021 and again in 2023.

India's monsoon works for four months and then stops, so almost every serious crop in this country stands on irrigation. RPSC does not ask you to analyse that. It asks project-to-river, project-to-state, scheme-to-year. Learn this chapter as a set of pairs and you will not lose a mark here.

Irrigation and Multipurpose Projects

Sources
- Tube wells ~45%
- Canals ~25%
- Other wells ~16%
- Tanks ~3%

North Indian dams
- Bhakra-Sutlej
- Tehri-Bhagirathi
- Rihand-Son basin
- Pong-Beas

Peninsular dams
- Hirakud-Mahanadi
- Nagarjuna Sagar-Krishna
- Tungabhadra
- Ukai-Tapi

Joint projects
- DVC 1948
- Kosi
- Gandak
- Chambal
- Bhakra-Nangal

Rajasthan link
- Indira Gandhi Canal
- Harike barrage
- Sardar Sarovar share
- Chambal valley

Schemes
- PMKSY 2015
- CADP 1974-75
- Atal Bhujal 2019
- Jal Jeevan 2019

Water harvesting
- Khadin
- Johad
- Zabo
- Ahar-pyne
- Kuhl

Interlinking
- NPP 1980
- NWDA 1982
- Ken-Betwa

Where India's irrigation water actually comes from

Before the projects, fix the arithmetic of sources. Net irrigated area is the land watered at least once in a year. Of that area, tube wells and other wells together do the heavy lifting - more than 60 per cent. Canals come second. Tanks, which every textbook talks about, are today a very small share. RPSC's favourite trick in this chapter is to inflate the tank figure and see if you notice.

Sources of irrigation by share of net irrigated area

Source	Approximate share	Where it dominates
Tube wells	About 45%	Punjab, Haryana, western UP, Bihar
Canals	About 25%	Northern plains, deltas of the east coast
Other wells	About 16%	Peninsular hard-rock areas
Other sources	About 11%	Scattered, includes lift and diversion
Tanks	About 3%	Peninsular plateau - Tamil Nadu, Telangana, AP

- Uttar Pradesh has the largest net irrigated area of any state.
- Punjab has the highest share of its net sown area irrigated - about 98 per cent.
- Rice takes the largest share of India's irrigated area; wheat is second.
- Wells plus tube wells together water over 60 per cent of the irrigated land.

Canal country vs tank country

Canal irrigation	Tank irrigation
Northern plains - flat land, perennial Himalayan rivers	Peninsular plateau - uneven rock, seasonal rivers
Alluvial soil, easy to dig, water holds level	Hard crystalline rock, natural hollows used as ponds
Punjab, Haryana, UP, Bihar, coastal deltas	Tamil Nadu, Telangana, Andhra Pradesh, Karnataka
About one-fourth of net irrigated area	About 3 per cent and falling

TRAP
Tanks irrigate about 3 per cent - not one-fourth. That inflated figure is a standard wrong statement in statement-based questions. Canals are the one-fourth.

What 'multipurpose' means, and the project that started it

A multipurpose project is one dam doing several jobs at once - irrigation, hydro power, flood control, navigation, fisheries, drinking water, sometimes tourism. Nehru called these dams the temples of modern India. The first one built after independence was the Damodar Valley Corporation, and it was consciously copied from America's Tennessee Valley Authority. Remember DVC as the debut.

- DVC set up in 1948 - first multipurpose river valley project of independent India.
- Model: Tennessee Valley Authority of the United States.
- Dams on the Barakar - Tilaiya and Maithon; on the Damodar - Konar and Panchet.
- It also runs the Durgapur barrage; the states served are Jharkhand and West Bengal.
- The Damodar was called the 'Sorrow of Bengal' before the project tamed its floods.

The order in which the big four came up - a favourite ordering question

DVC created, 1948 → Hirakud completed, 1957 → Bhakra dedicated to the nation, 1963 → Farakka barrage commissioned, 1975

The big dams - river, state and the one fact that identifies each

This is the core table of the chapter. RPSC asks it as project-to-river or project-to-state matching, and sometimes as a negative question - 'which pair is NOT correctly matched'. Do not try to remember heights for all of them. Remember one identifying label per dam, because that is what the option strings carry.

Project - river - state (set 1)

Project	River	State(s)
Bhakra	Sutlej	Himachal Pradesh (Bilaspur)
Nangal	Sutlej, below Bhakra	Punjab
Tehri	Bhagirathi and Bhilangana	Uttarakhand
Rihand	Rihand, a Son tributary	Uttar Pradesh (Sonbhadra)
Pong (Maharana Pratap Sagar)	Beas	Himachal Pradesh
Ranjit Sagar (Thein)	Ravi	Punjab-J&K border
Nathpa Jhakri	Satluj	Himachal Pradesh
Farakka barrage	Ganga	West Bengal
Hirakud	Mahanadi	Odisha
Sardar Sarovar	Narmada	Gujarat

Project - river - state (set 2)

Project	River	State(s)
Nagarjuna Sagar	Krishna	Telangana / Andhra Pradesh
Tungabhadra	Tungabhadra	Karnataka and Andhra Pradesh
Sriramsagar	Godavari	Telangana
Mettur (Stanley)	Kaveri	Tamil Nadu
Idukki (arch dam)	Periyar	Kerala

Project	River	State(s)
Koyna	Koyna	Maharashtra
Ukai	Tapi	Gujarat
Bansagar	Son	MP, UP and Bihar
Gandhi Sagar	Chambal	Madhya Pradesh
Rana Pratap Sagar, Jawahar Sagar, Kota barrage	Chambal	Rajasthan

- Tehri - 260.5 m, the tallest dam in India.
- Bhakra - 226 m, the highest straight gravity dam in India; reservoir Gobind Sagar.
- Hirakud - with its dykes about 25.8 km, among the longest dams in the world; built 1957.
- Nagarjuna Sagar - one of the largest masonry dams; canals Jawahar (right bank) and Lal Bahadur (left bank).
- Rihand - reservoir Govind Ballabh Pant Sagar, one of India's largest artificial lakes.
- Farakka, 1975 - diverts Ganga water into the Bhagirathi-Hooghly to flush silt and keep Kolkata port navigable.
- Sardar Sarovar - the Narmada main canal is among the world's largest irrigation canals.

TRAP

Two Bhakra traps in one box. First, Gobind Sagar is Bhakra's reservoir on the Sutlej, while Govind Ballabh Pant Sagar is Rihand's in Uttar Pradesh - the names look almost identical in an options list. Second, Bhakra is on the Sutlej, never on the Beas; 'Bhakra - Beas' is the standard planted wrong pair. The Beas dam is Pong.

Joint and inter-state projects

Rivers do not respect state lines, so several projects are run jointly. RPSC likes these because the answer is a list of states, which is easy to scramble in an option. Two of them cross an international border - Kosi and Gandak both involve Nepal, the Kosi being the river called the 'Sorrow of Bihar'. Note also that Rajasthan appears twice here, as a partner in Bhakra-Nangal and as a beneficiary of Sardar Sarovar.

Project to participating states

Project	Participating states / countries
Bhakra-Nangal	Punjab, Haryana, Rajasthan, Himachal Pradesh
Damodar Valley	Jharkhand and West Bengal
Kosi	Bihar and Nepal
Gandak	Bihar, Uttar Pradesh and Nepal
Chambal valley	Madhya Pradesh and Rajasthan
Tungabhadra	Karnataka and Andhra Pradesh
Parambikulam-Aliyar	Tamil Nadu and Kerala
Bansagar	Madhya Pradesh, Uttar Pradesh and Bihar
Sardar Sarovar	Gujarat, Madhya Pradesh, Maharashtra, Rajasthan

- The Chambal valley project has four structures - Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar and the Kota barrage.
- Nangal Hydel Channel feeds the Bhakra main line canal.
- Kosi is Bihar plus Nepal; Gandak adds Uttar Pradesh to the same pair - the shorter name has the shorter list.

The Indira Gandhi Canal – the one project Rajasthan is asked about

This is the longest canal in India and the single most examined irrigation work in the whole RAS paper. It was built to carry Punjab's rivers into the Thar. It was originally called the Rajasthan Canal and was renamed after Indira Gandhi. Learn the water's route from source to tail – questions are built on exactly that sequence.

The route of the water

Sutlej and Beas meet → Harike barrage, Punjab → Rajasthan Feeder, 204 km → Indira Gandhi main canal, 445 km → Tail at Mohangarh, Jaisalmer

- Off-takes from the Harike barrage, just below the Sutlej-Beas confluence.
- It carries the waters of the Sutlej and the Beas, not the Ganga system.
- Rajasthan Feeder 204 km plus main canal 445 km – the longest canal in India.
- Built in two stages; several lift canals take water uphill from it, including Kanwar Sen (Gharsana), Pannalal Barupal and Chaudhary Kumbha Ram.
- The command covers the north-western districts, from Sriganganagar and Hanumangarh down to Bikaner, Jaisalmer and Barmer.
- Its criticism is waterlogging and rising salinity in parts of the command.

ASKED IN RAS

Asked in RAS Pre 2023 - the Indira Gandhi Canal takes off from which barrage? Answer: Harike. If you remember nothing else about this canal, remember Harike.

TRAP

The main canal does not end at Sriganganagar. It runs south-west to Mohangarh in Jaisalmer. Sriganganagar sits on the older Gang Canal system, commissioned in 1927.

Schemes - name, year, and the one line that identifies each

Scheme questions are pure recall. The examiner picks the launch year, the ministry or the component list. Learn those three things for each scheme and skip the rest of the brochure.

Irrigation and water schemes

Scheme	Year	Identifier
Command Area Development Programme	1974-75	Now CAD and Water Management, under PMKSY
National Water Policy	1987	Revised in 2002 and again in 2012
National Water Development Agency	1982	Executes the National Perspective Plan of 1980
PMKSY	1 July 2015	Motto 'Har Khet Ko Pani'
Atal Bhujal Yojana	25 December 2019	Groundwater, Ministry of Jal Shakti
Jal Jeevan Mission	2019	'Har Ghar Jal' - household tap connections

- PMKSY 2021-26 phase outlay: Rs 93,068.56 crore.

- Its three components are AIBP, Har Khet Ko Pani, and Watershed Development.
- 'Per Drop More Crop' - drip and sprinkler - was a PMKSY component till 2021-22.
- From 2022-23 Per Drop More Crop is implemented under the Rashtriya Krishi Vikas Yojana.
- The Micro Irrigation Fund sits with NABARD.
- The National Water Mission under NAPCC targets a 20 per cent improvement in water-use efficiency.

Atal Bhujal Yojana - the most repeated scheme in this chapter

- Launched 25 December 2019, on Vajpayee's birth anniversary; Ministry of Jal Shakti.
- Outlay Rs 6,000 crore - half a World Bank loan, half the Central share.
- It is demand-side and community-led: gram panchayats prepare water security plans.
- It covers only seven water-stressed states - Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh.

ASKED IN RAS

Asked in RAS Pre 2021 - which ministry implements Atal Bhujal Yojana? Answer: Jal Shakti. Asked again in RAS Pre 2023 as a statement question, where the wrong statement was 'it is being implemented in all the states of India'. Two papers, one scheme.

TRAP

Punjab is NOT in the Atal Bhujal list, even though its groundwater is among the worst in India. The seven states are Gujarat, Haryana, Karnataka, MP, Maharashtra, Rajasthan, UP. Mnemonic: GH-KMM-RU.

Groundwater, traditional harvesting and river interlinking

India draws more groundwater than any country on earth, so the Central Ground Water Board's Dynamic Assessment has become a standard source of figures. Learn the four headline numbers of the 2024 assessment and the safe-versus-over-exploited split. The examiner will change one digit and ask you to spot it.

- Dynamic Ground Water Resource Assessment 2024: annual recharge 446.90 BCM.
- Annual extractable resource 406.19 BCM; actual extraction 245.64 BCM.
- Stage of extraction for the country: 60.47 per cent.
- Of 6,746 assessment units, 4,951 are 'safe' - about 73.4 per cent.
- 751 units are 'over-exploited' (11.1 per cent), 206 'critical' and 711 'semi-critical'.

Traditional water harvesting - a matching question waiting to happen

Structure to state

Structure	State / region
Khadin and tanka	Rajasthan (khadin - Jaisalmer)
Johad	Alwar, Rajasthan
Kuhl	Himachal Pradesh
Zabo	Nagaland
Apatani	Arunachal Pradesh
Ahar-pyne	Bihar
Eri tanks	Tamil Nadu
Surangam	Kerala
Phad	Maharashtra
Bamboo drip irrigation	Meghalaya

ASKED IN RAS

Asked in RAS Pre 2024 as a match-the-following - khadin/Rajasthan, zabo/Nagaland, ahar-pyne/Bihar, kuhl/Himachal. Learn all ten pairs above; RPSC rotates which four it puts in the list.

River interlinking

- National Perspective Plan framed in 1980; executed by the National Water Development Agency, set up 1982.
- Ken-Betwa is the first river-link project actually taken up.
- It involves Madhya Pradesh and Uttar Pradesh; the storage is the Daudhan dam on the Ken, in Panna.
- Namami Gange is the umbrella Ganga rejuvenation programme - do not confuse it with a link project.

CURRENT LINK

Current link, as of September 2026: the foundation stone of the Ken-Betwa Link Project was laid on 25 December 2024 at Khajuraho and construction of the Daudhan dam is under way. The standing controversy is submergence in the core area of the Panna Tiger Reserve. Expect this either as 'India's first river-link project' or paired with Panna.

REVISION RECAP

1. Wells and tube wells irrigate over 60 per cent of India's net irrigated area; canals about 25 per cent; tanks only about 3 per cent.
2. UP has the largest net irrigated area; Punjab has the highest irrigated share of its sown area (about 98 per cent); rice takes the largest share of irrigated land.
3. DVC, 1948 - first multipurpose project of independent India, on the TVA model; Tilaiya and Maithon on the Barakar, Konar and Panchet on the Damodar.
4. Order to remember: DVC 1948, Hirakud 1957, Bhakra 1963, Farakka 1975.

5. Tehri is the tallest dam (260.5 m); Bhakra the highest straight gravity dam (226 m); Hirakud among the longest; Nagarjuna Sagar among the largest masonry dams.
6. Bhakra-Sutlej-Bilaspur, reservoir Gobind Sagar. Rihand-Sonbhadra, reservoir Govind Ballabh Pant Sagar. Keep the two reservoirs apart.
7. Farakka, 1975, diverts water into the Bhagirathi-Hooghly for Kolkata port.
8. Nepal appears in two projects - Kosi (Bihar) and Gandak (Bihar plus UP).
9. Chambal valley = MP plus Rajasthan; four structures - Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar, Kota barrage.
10. Sardar Sarovar on the Narmada benefits Gujarat, MP, Maharashtra and Rajasthan.
11. Indira Gandhi Canal: Harike barrage, Rajasthan Feeder 204 km, main canal 445 km, tail at Mohangarh in Jaisalmer; India's longest canal.
12. PMKSY - 1 July 2015, 'Har Khet Ko Pani', components AIBP, HKKP and Watershed; Per Drop More Crop moved to RKVY from 2022-23.
13. Atal Bhujal Yojana - 25 December 2019, Jal Shakti, Rs 6,000 crore, half World Bank, seven states, gram panchayat water security plans.
14. Command Area Development Programme began in 1974-75; National Water Policy first framed in 1987.
15. Groundwater 2024: recharge 446.90 BCM, extraction 245.64 BCM, stage 60.47 per cent, 751 over-exploited units.
16. Water harvesting pairs: khadin/tanka Rajasthan, johad Alwar, kuhl Himachal, zabo Nagaland, apatani Arunachal, ahar-pyne Bihar, eri Tamil Nadu, surangam Kerala, phad Maharashtra, bamboo drip Meghalaya.
17. National Perspective Plan 1980, NWDA 1982, Ken-Betwa the first link taken up - Daudhan dam, MP and UP, foundation stone 25 December 2024.

IGOS

Agriculture and Crops of India

About 1 question a paper from the Indian Geography slot, plus spillover into Indian Economy and Current Affairs through MSP, PM-KISAN and crop insurance. Seasons, soils and the revolutions are the three blocks that repeat.

Agriculture supports nearly half of India's workforce but contributes only about 18 per cent of gross value added - that gap is the whole story of Indian farming. For the exam, though, this chapter is a matching chapter: crop to season, crop to state, soil to region, revolution to product, scheme to year.

Agriculture and Crops of India

Seasons
- Kharif - monsoon sown
- Rabi - winter sown
- Zaid - short summer

Cereals
- Rice
- Wheat
- Bajra
- Jowar
- Maize

Cash crops
- Sugarcane
- Cotton
- Jute
- Oilseeds

Plantation
- Tea-Assam
- Coffee-Karnataka
- Rubber-Kerala

Soils
- Alluvial
- Black/regur
- Red-yellow
- Laterite
- Arid
- Saline

Revolutions
- Green-foodgrain
- White-milk
- Blue-fish
- Yellow-oilseeds

Price and market
- CACP
- MSP 22 crops
- FRP sugarcane
- FCI
- e-NAM

Schemes
- PM-KISAN
- PMFBY
- Soil Health Card
- NFSM
- RKVY
- PKVY

The three cropping seasons

India's farming calendar is set by the monsoon, not by the English calendar. Kharif is sown when the south-west monsoon arrives and harvested when it leaves. Rabi is sown into the cooling soil after the monsoon and harvested in spring. Zaid is the short gap crop between them. Almost every year some paper somewhere asks you to pick the odd crop out of a season list, so learn the crop lists cold.

Kharif vs Rabi

Kharif	Rabi
Sown June-July, with the onset of the south-west monsoon	Sown October-December, after the monsoon withdraws
Harvested September-October	Harvested April-June
Needs high temperature and heavy rain	Needs a cool growing season and bright sunshine at ripening
Rice, jowar, bajra, maize, cotton, jute, groundnut, tur, soybean	Wheat, barley, gram, mustard, peas, linseed
Rain-fed over much of the country	Depends heavily on irrigation and winter western disturbances

- Zaid runs in the short season between the rabi harvest and the kharif sowing.
- Zaid crops: watermelon, muskmelon, cucumber, summer moong and fodder crops.
- Gram and mustard are rabi, not zaid - a standard planted error in statement questions.
- Barley is rabi: it is planted inside kharif lists as the odd one out in 'which is NOT a kharif crop'.

Foodgrains - conditions and leading states

For each major crop the examiner wants two things: the growing condition (rainfall and temperature) and the leading producer state. Do not chase the second and third ranks - those move with the year. Fix the leader, and know which leaders are stable.

- Rice - kharif, over 100 cm rainfall, about 25 degrees C; grown with canal and tube-well water in Punjab and Haryana where rainfall is low.
- West Bengal grows three rice crops - aus, aman and boro.
- Punjab has the highest rice yield per hectare among the major states.
- Wheat - rabi, 50-75 cm rainfall, cool growing season with bright sunshine at ripening; Punjab has the highest wheat productivity.
- Uttar Pradesh is the leading producer of wheat.
- Rajasthan leads in bajra and in barley; Maharashtra leads in jowar.

Crop to leading producer state - the stable ones

Crop	Leading producer state
Wheat	Uttar Pradesh
Sugarcane	Uttar Pradesh
Jute	West Bengal
Tea	Assam
Coffee	Karnataka
Rubber	Kerala
Rapeseed-mustard, bajra, barley	Rajasthan
Soybean, gram, total pulses	Madhya Pradesh
Groundnut	Gujarat
Jowar, onion	Maharashtra
Potato, mango	Uttar Pradesh

Crop	Leading producer state
Mulberry silk	Karnataka

TRAP

Rice and cotton ranks move. Cotton is the classic contested one: Gujarat is usually the largest producer, but the largest area under cotton is in Maharashtra, and the two states trade the production lead in bad years. If an option pair asks for cotton, read whether it says area or production.

- India is the world's largest producer of milk, pulses, jute and spices.
- India is the second largest producer of rice, wheat, sugarcane, groundnut, fruits and vegetables.
- China is first in wheat and rice - so 'India is the largest producer of wheat' is always a wrong statement.

Cash crops and plantation crops

Cash crops carry the trade and the history, so they come with dates and nicknames - and RPSC loves both. Sugarcane is the one to be careful with, because it is the only major crop that does not get an MSP.

- Sugarcane - Uttar Pradesh leads in both area and production; sugar recovery is higher in the tropical belt of Maharashtra and Karnataka.
- Sugarcane gets a Fair and Remunerative Price (FRP), not an MSP.
- Jute, the 'golden fibre' - needs 150-200 cm rainfall, grows in the Ganga-Brahmaputra delta.
- India's first jute mill was set up at Rishra, on the Hooghly, in 1855.
- Tea was introduced in Assam in the 1830s; other belts are Darjeeling and the Nilgiris.
- Coffee - Arabica and Robusta - began in the Baba Budan hills of Karnataka.

- Rubber needs over 200 cm rainfall; Kerala dominates.

ASKED IN RAS

Asked in RAS Pre 2024 in the world geography slot - which regions practise Mediterranean agriculture. RPSC clearly tests agricultural typology, not just crop lists. The Indian counterpart of that instinct is the shifting-cultivation names question in the next section.

Soils of India

Soil questions in RAS are always about two things - how the soil formed and what it is rich or poor in. Black soil comes from lava, laterite from leaching, red from crystalline rock. Learn the formation verb for each and half the question answers itself.

Soil to region

Soil	Typical region	Key trait
Alluvial	Northern plains and the deltas	Largest area of any Indian soil
Black (regur)	Deccan plateau - Maharashtra, MP, Gujarat	From basaltic lava; self-ploughing cracks
Red and yellow	Eastern and southern Deccan, Chotanagpur	From crystalline igneous rock; iron oxide colour
Laterite	Malabar coast, Western Ghats, Odisha, north-east	Intense leaching; used as building brick
Arid / desert	Western Rajasthan	Sandy, saline, low organic matter
Forest and mountain	Himalayan slopes	Thin, acidic on higher slopes
Saline and alkaline	Parts of Punjab, Haryana, Gujarat, Rajasthan	Salt crust from over-irrigation and evaporation

Soil	Typical region	Key trait
Peaty and marshy	Kerala backwaters, coastal Odisha, Sundarbans	High organic matter, waterlogged

- Black soil is rich in lime, iron, magnesia and potash - and poor in nitrogen, phosphorus and humus.
- It swells when wet and cracks deeply in summer, which aerates it; hence 'self-ploughing'.
- Laterite is poor in fertility but suits tea, coffee, cashew and rubber.
- Red soils are deficient in nitrogen, phosphorus and humus, like black soil.

TRAP

'Black soil is rich in nitrogen, phosphorus and organic matter' is FALSE and it is the most planted wrong statement in this chapter. Rich in lime-iron-magnesia-potash; poor in N, P and humus. Same for laterite - it is not rich in humus.

YAAD RAKHO

Formation verbs: black = lava cooled, laterite = leached, red = weathered crystalline rock, alluvial = deposited by rivers. Lava, leach, rock, river.

Land use, agro-climatic zones and shifting cultivation

Two sets of numbers get asked here - how the country's land is used, and how many agro-climatic zones there are. The zone count has three different answers depending on who divided the country, so read the question stem carefully.

- Net sown area is about 46 per cent of the reporting area - the single largest land use in India.

- Cropping intensity = gross cropped area divided by net sown area.
- Planning Commission: 15 agro-climatic zones.
- ICAR / NARP: 127 agro-climatic zones.
- NBSS and LUP: 20 agro-ecological regions.
- Agriculture and allied sectors give roughly 18 per cent of gross value added but employ nearly half the workforce.

Shifting cultivation – the local names

Name	Region
Jhum	North-eastern India
Podu	Andhra Pradesh and Odisha
Bewar and dahiya	Madhya Pradesh
Kumari	Kerala
Walra	Bhils of southern Rajasthan

ASKED IN RAS

Asked in RAS Pre 2013 and again in 2018 – the slash-and-burn cultivation of the Bhils of Rajasthan. Answer: Walra. Because it is the Rajasthan entry in a national list, RPSC returns to it. Learn all five names, not just walra.

The revolutions

Every colour revolution is simply a product plus a programme plus a person. The Green Revolution is the one with the fullest story, so it gets asked in detail; the rest are asked as one-word matching. Do not overload yourself – one product per colour is enough.

Revolution to product

Revolution	Product	Anchor fact
Green	Foodgrains, especially wheat	1966-67, HYV Mexican dwarf wheat; M. S. Swaminathan
White	Milk	Operation Flood 1970, NDDB, Verghese Kurien, Anand pattern
Blue	Fisheries	Inland and marine fish production
Yellow	Oilseeds	Technology Mission on Oilseeds, 1986
Silver	Eggs and poultry	Poultry expansion
Pink	Prawns and meat processing	Not to be confused with silver
Round	Potato	Short and easy to forget
Grey	Fertilisers	Input side, not output

How the Green Revolution actually ran

Mexican dwarf HYV wheat of Norman Borlaug imported → Package of seed, fertiliser and assured irrigation, 1966-67 → Punjab, Haryana and western Uttar Pradesh gain most → Wheat output jumps; India moves towards self-sufficiency → IR-8 'miracle rice' from IRRI extends it to rice

- M. S. Swaminathan is called the father of India's Green Revolution.
- Norman Borlaug bred the dwarf wheat varieties in Mexico.
- IR-8 is a rice variety from the International Rice Research Institute in the Philippines - not a wheat variety.
- Operation Flood, 1970, built on Anand-pattern cooperatives under NDDB.

TRAP

The single commonest wrong statement about the Green Revolution: 'it began with the IR-8 variety of wheat'. IR-8 is rice, and Philippine. The wheat came from Mexico.

Prices, marketing and the schemes

The price chain is short and worth memorising as a sequence, because questions attack any one link in it. A body recommends, a cabinet committee announces, an agency procures, and a network distributes.

How a support price reaches a farmer

CACP (set up 1965) recommends the price → CCEA announces the MSP → Food Corporation of India (1965) procures → FCI stores against buffer norms → Grain moves into the public distribution system

- MSP is announced for 22 crops, plus the FRP for sugarcane.
- Since 2018-19 MSP is fixed at not less than 1.5 times the A2+FL cost.
- A2+FL = paid-out costs plus the imputed value of family labour.
- e-NAM, the electronic National Agriculture Market, was launched on 14 April 2016.
- e-NAM is run by the Small Farmers' Agribusiness Consortium (SFAC) and links APMC mandis online.

Scheme to year

Scheme	Year	Identifier
Kisan Credit Card	1998	Short-term credit to farmers

Scheme	Year	Identifier
National Food Security Mission	2007-08	Rice, wheat, pulses, coarse cereals
Rashtriya Krishi Vikas Yojana	2007	State-led agriculture planning
Soil Health Card	19 February 2015	Launched from Suratgarh, Rajasthan
PMFBY	Kharif 2016	Replaced NAIS and MNAIS
e-NAM	14 April 2016	Run by SFAC
Operation Greens	2018	Tomato, onion, potato - TOP
PM-KISAN	Announced 24 Feb 2019, effective 1 Dec 2018	Rs 6,000 a year

- PMFBY farmer premium: 2 per cent for kharif, 1.5 per cent for rabi food and oilseed crops, 5 per cent for commercial and horticultural crops.
- PMFBY was made voluntary for loanee farmers in 2020.
- PM-KISAN pays Rs 6,000 a year in three equal instalments of Rs 2,000.
- Paramparagat Krishi Vikas Yojana promotes organic farming; Sikkim became India's first fully organic state in 2016.
- The National Agriculture Policy, 2000 set a 4 per cent annual growth target for the sector.

YAAD RAKHO

Soil Health Card was launched from Suratgarh in Rajasthan. Any national scheme with a Rajasthan launch site is a gift to an RPSC paper-setter - remember the place, not just the date.

Crop research institutes

Institute	Headquarters
National Rice Research Institute (formerly CRRI)	Cuttack
Indian Institute of Sugarcane Research	Lucknow

Institute	Headquarters
Central Potato Research Institute	Shimla
Indian Institute of Pulses Research	Kanpur
Indian Institute of Wheat and Barley Research	Karnal
National Research Centre on Seed Spices	Ajmer
National Research Centre on Camel	Bikaner
Central Institute for Arid Horticulture	Bikaner

ASKED IN RAS

Asked in RAS Pre 2016, 2018 and 2023 - the national research centres located in Rajasthan. Seed Spices at Ajmer, Camel at Bikaner, Arid Horticulture at Bikaner. Three papers, the same three institutes. Learn the city with each.

CURRENT LINK

Current link, as of September 2026: PM-KISAN crossed its 22nd instalment, released in March 2026 from Guwahati, and a further instalment followed in mid-2026. The instalment number changes every four months, so never learn it - learn Rs 6,000 a year in three instalments of Rs 2,000, which does not change.

REVISION RECAP

1. Kharif is sown with the monsoon (June-July) and harvested September-October; rabi is sown October-December and harvested April-June; zaid fills the gap.
2. Kharif = rice, jowar, bajra, maize, cotton, jute, groundnut, tur, soybean. Rabi = wheat, barley, gram, mustard, peas, linseed.
3. Rice needs over 100 cm rain and about 25 degrees C; West Bengal grows aus, aman and boro; Punjab has the highest rice and wheat yields.
4. Stable leaders: wheat and sugarcane UP, jute West Bengal, tea Assam, coffee Karnataka, rubber Kerala, groundnut Gujarat, soybean and pulses MP, mustard/bajra/barley Rajasthan.

5. India is the world's largest producer of milk, pulses, jute and spices; second in rice, wheat, sugarcane and groundnut.
6. Sugarcane gets FRP, not MSP; UP leads in area and production, Maharashtra and Karnataka in recovery.
7. First jute mill at Rishra, 1855; first cotton textile mill at Mumbai, 1854 - keep the pair straight.
8. Alluvial soil covers the largest area; black soil comes from Deccan lava and is poor in nitrogen, phosphorus and humus; laterite is formed by intense leaching.
9. Planning Commission 15 agro-climatic zones, ICAR/NARP 127, NBSS and LUP 20 agro-ecological regions; net sown area about 46 per cent.
10. Shifting cultivation: jhum north-east, podu AP-Odisha, bewar/dahiya MP, kumari Kerala, walra among the Bhils of Rajasthan.
11. Green Revolution 1966-67 with Mexican dwarf wheat; Swaminathan the father in India, Borlaug the breeder; IR-8 is rice from IRRI.
12. White = milk (Operation Flood 1970, Kurien, NDDB), Yellow = oilseeds (Technology Mission 1986), Blue = fish, Silver = eggs, Pink = prawns, Round = potato, Grey = fertilisers.
13. CACP (1965) recommends MSP for 22 crops; since 2018-19 it is at least 1.5 times A2+FL; FCI (1965) procures and stores.
14. e-NAM launched 14 April 2016 under SFAC; PMFBY from kharif 2016 with 2 / 1.5 / 5 per cent farmer premiums.
15. PM-KISAN - Rs 6,000 a year in three instalments of Rs 2,000; announced 24 February 2019, effective from 1 December 2018.
16. Soil Health Card launched 19 February 2015 from Suratgarh, Rajasthan; Sikkim India's first fully organic state, 2016.
17. Rajasthan's national research centres: Seed Spices Ajmer, Camel Bikaner, Arid Horticulture Bikaner - asked in 2016, 2018 and 2023.

IG06

Minerals, Industries and Transport of India

1 to 2 questions a paper, and the highway corridor block alone has appeared in five separate papers. Minerals feed the Rajasthan Geography slot as well. If you are short of time in this subject, do this chapter first and the highway section within it first of all.

This is the highest-yield chapter in Indian Geography, and it contains the single most repeated geography topic in the whole RAS paper - the national highway corridors and their crossing at Jhansi, asked in 2013, 2016, 2021, 2023 and 2024. Everything else here is mineral-to-place and plant-to-collaborator matching.

Minerals, Industries and Transport

Iron ore belts
- Odisha-Jharkhand
- Durg-Bastar-Chandrapur
- Bellary-Chitradurga
- Maharashtra-Goa

Other metals
- Manganese-Balaghat
- Bauxite-Panchpatmali
- Copper-Malanjkhand
- Lead-zinc-Rampura Agucha

Energy
- Jharia coking coal
- Neyveli lignite
- Mumbai High
- Rawatbhata
- Kudankulam

Steel plants
- Jamshedpur-Tata
- Bhilai-USSR
- Rourkela-W Germany
- Durgapur-UK
- Vizag-shore

Industrial regions
- Hugli oldest
- Mumbai-Pune
- Chotanagpur
- Gujarat
- Bengaluru-TN

Highways
- Golden Quadrilateral
- North-South NH-44
- East-West NH-27
- Cross at Jhansi

Rail and water
- 19 zones
- DFC East and West
- NW-1 Ganga
- NW-2 Brahmaputra

Ports
- 12 major ports
- JNPT container
- Kolkata riverine
- Vadhavan new

Metallic minerals – the belts and who leads

Mineral questions come in one shape: a mine or a belt on one side, a state on the other. So learn the place name first and the geology never. One caution – for several minerals the largest producer and the largest reserve holder are different states, and RPSC builds assertion-reason and NOT-matched questions on exactly that gap.

Mineral – belt or mine – state

Mineral	Belt / mine	State
Iron ore	Badampahar, Gua, Noamundi	Odisha and Jharkhand
Iron ore	Bailadila, Dalli-Rajhara	Chhattisgarh (Durg-Bastar-Chandrapur belt)
Iron ore	Kudremukh	Karnataka (Bellary-Chitradurga-Tumakuru belt)
Manganese	Balaghat	Madhya Pradesh
Manganese	Nagpur-Bhandara	Maharashtra
Copper	Malanjkhand (Balaghat)	Madhya Pradesh
Copper	Khetri	Rajasthan
Copper	Singhbhum	Jharkhand
Bauxite	Panchpatmali (Koraput), Kalahandi	Odisha
Bauxite	Lohardaga	Jharkhand
Lead-zinc	Rampura Agucha, Zawar, Rajpura-Dariba	Rajasthan
Gold	Hutti and Kolar	Karnataka
Chromite	Sukinda valley	Odisha

- Odisha is the largest producer of iron ore in India.
- Bailadila haematite (Dantewada, Chhattisgarh) is exported to Japan through Visakhapatnam.

- Kudremukh magnetite (Karnataka) moved out through New Mangalore.
- Manganese - Madhya Pradesh leads production (about 31 per cent as per the Indian Bureau of Mines); Odisha holds the largest reserves (about 34 per cent).
- Copper - Madhya Pradesh is the largest producer; Rajasthan holds over half the country's copper reserves.
- Lead-zinc - almost the entire Indian output is from Rajasthan, worked by Hindustan Zinc.
- Aluminium - Odisha leads; NALCO refines at Damanjodi and smelts at Angul, BALCO is at Korba, HINDALCO at Renukoot.
- Kolar Gold Fields closed in 2001; Hutti still works, and Ramagiri is in Andhra Pradesh.
- Three copper belts, west to east: Khetri (Rajasthan), Malanjkhand (MP), Singhbhum (Jharkhand).

TRAP

Manganese is the classic assertion-reason trap. 'Odisha is the leading producer of manganese' is FALSE - Madhya Pradesh is. But 'Odisha holds the largest manganese reserves' is TRUE. Producer MP, reserves Odisha. Copper works the same way: producer MP, reserves Rajasthan.

Non-metallic and atomic minerals

A short section but a dense one - almost every entry is a one-line pair, and several of them are in Rajasthan, which is why RPSC keeps coming back to them. Note the split that runs through this section too: Rajasthan is the largest producer of limestone, but Karnataka holds the largest reserves.

Non-metallic and atomic minerals

Mineral	Chief location	State / note
Limestone	Widespread	Rajasthan largest producer (about 22%); Karnataka largest reserves
Rock phosphate	Jhamarkotra	Rajasthan (Udaipur) - India's chief mine
Gypsum	Nagaur, Bikaner	Rajasthan
Mica	Koderma-Giridih	Jharkhand; also Nellore (AP) and Bhilwara
Diamond	Panna	Madhya Pradesh
Uranium	Jaduguda	Jharkhand - India's first uranium mine
Uranium	Tummalapalle	Andhra Pradesh
Chromite	Sukinda	Odisha - nearly all of India's output

Energy resources - coal, oil, gas and nuclear

Coal first. India's coal is overwhelmingly Gondwana coal, found in the peninsular river valleys; Tertiary coal is younger, poorer and sits in the north-east. The examiner's favourite pair here is 'oldest coalfield' and 'coking coal coalfield' - they are two different fields. Then move to oil, where Rajasthan has a big stake.

- About 96 per cent of India's coal reserves are Gondwana coal.
- Jharia is the largest coalfield and the chief source of prime coking coal.
- Raniganj, where mining began in 1774, is the oldest coalfield.
- Other fields: Bokaro, Karanpura, Korba, Singrauli, Talcher.
- Jharkhand has the largest coal reserves; Odisha has led production in recent years.
- Lignite - Neyveli (Tamil Nadu) is the largest field; Kapurdi-Jalipa and Barsingsar are in Rajasthan; Panandhro in Gujarat.

- Mumbai High, found 1974 and drilled by the rig Sagar Samrat, is India's largest oil field.
- Rajasthan's Barmer basin - the Mangala field, discovered 2004 - makes it the leading onshore crude producer.
- Gujarat has Ankleshwar and Kalol; Assam has Digboi, Naharkatiya and Moran.
- Digboi, 1901, is India's oldest refinery; Jamnagar in Gujarat is the world's largest refining complex.
- The HPCL Rajasthan Refinery is at Pachpadra in Balotra, erstwhile Barmer district.
- Natural gas moves through GAIL's Hazira-Vijaipur-Jagdishpur (HVJ) pipeline.
- Hydel power comes from the multipurpose dams of the previous chapter; among renewables, Bhadla in Rajasthan is among the world's largest solar parks.

ASKED IN RAS

Rajasthan's fossil fuels have been asked twice over. RAS Pre 2024 - which lignite field lies in Rajasthan? Answer: Kapurdi-Jalipa (Barmer); Barsingsar and Palana are the other names. RAS Pre 2013 and 2018 - the Mangala oil field lies in which basin? Answer: the Barmer basin, with Bhagyam and Aishwariya as the companion fields.

Nuclear power plant - state - identifier

Plant	State	Identifier
Tarapur	Maharashtra	Oldest, commissioned 1969
Rawatbhata	Rajasthan	Built with Canadian help
Kalpakkam	Tamil Nadu	Fast breeder programme site
Kudankulam	Tamil Nadu	Largest, Russian-aided
Kakrapar	Gujarat	India's first indigenous 700 MWe PHWR
Narora	Uttar Pradesh	On the Ganga canal

Plant	State	Identifier
Kaiga	Karnataka	Western Ghats site

TRAP
Two planted errors sit together here. 'Rajasthan has the largest coal reserves' is false - Rajasthan has lignite, Jharkhand has the coal. And Neyveli is a lignite-thermal complex, not a nuclear station; it is the standard wrong option in 'which is NOT a nuclear power station'.

Iron and steel, and the other big industries

Steel plants are asked in exactly one way: which country helped build which plant. Learn the four SAIL collaborations as a block, then add the two that stand apart - Tata's private plant at Jamshedpur on the Subarnarekha, India's first modern steel plant, and Visakhapatnam, the only shore-based integrated plant.

Steel plant - state - collaboration

Plant	State	Built with
Jamshedpur (TISCO)	Jharkhand	Private, Tata - founded 1907, production from 1912
Bhilai	Chhattisgarh	USSR
Bokaro	Jharkhand	USSR
Rourkela	Odisha	West Germany
Durgapur	West Bengal	United Kingdom
Visakhapatnam (RINL)	Andhra Pradesh	Only shore-based integrated plant

- Order of production: Jamshedpur 1912, Bhilai 1959, Bokaro early 1970s, Visakhapatnam 1992.
- The first cotton textile mill was set up at Mumbai in 1854; the first jute mill at Rishra on the Hooghly in 1855.
- Ahmedabad is the 'Manchester of India'; Coimbatore the 'Manchester of South India'.
- Sugar - Uttar Pradesh has the most mills and the largest cane area; Maharashtra has higher recovery.
- Cement - the first plant was set up at Chennai in 1904; India is the world's second largest cement producer.
- Sindri, 1951, was India's first large fertiliser plant.
- Chennai is the 'Detroit of India'; other auto hubs are Gurugram-Manesar, Pune-Chakan, Sanand and Pithampur.
- Bengaluru is the 'Silicon Valley of India'.

YAAD RAKHO

Two Russians, one German, one Briton: Bhilai and Bokaro from the USSR, Rourkela from West Germany, Durgapur from the United Kingdom. The D plant is NOT Deutschland - that is exactly the slip the option list is waiting for.

Industrial regions and corridors

India is usually described as having eight major industrial regions. Learn each with its signature industry, because the matching question uses the industry, not the geography.

Industrial region - characteristic industry

Region	Signature industry
Hugli (the oldest)	Jute
Mumbai-Pune	Cotton textiles, engineering, films
Chotanagpur	Iron and steel, heavy metallurgy
Gujarat	Cotton textiles and petrochemicals
Vishakhapatnam-Guntur	Shipbuilding and petroleum refining
Bengaluru-Tamil Nadu	Electronics, engineering, textiles
Gurugram-Delhi-Meerut	Light engineering, electronics, automobiles
Kollam-Thiruvananthapuram	Coir, rubber, cashew, sugar, aluminium

- Dedicated Freight Corridors: Western DFC from Dadri (UP) to JNPT Mumbai, about 1,506 km, with Japanese assistance.
- Eastern DFC from Sahnewal (Ludhiana) to Sonnagar (Bihar), about 1,337 km, with World Bank assistance.
- The Delhi-Mumbai Industrial Corridor runs along the Western DFC and is Japan-aided.
- About 39 per cent of the DMIC alignment lies in Rajasthan - the state's biggest industrial hook.
- Other corridors: Chennai-Bengaluru, Amritsar-Kolkata, Vizag-Chennai.
- Sagarmala (2015) is port-led development; Make in India is 2014; PM Gati Shakti is 2021 for multimodal connectivity.
- Two planted errors: Kollam-Thiruvananthapuram rests on coir, rubber and cashew, NOT iron and steel; and Chotanagpur is Jharkhand, not Chhattisgarh.

National highways and the corridors - the most repeated topic in RAS geography

Give this section a whole revision slot to itself. RPSC has asked from it in 2013, 2016, 2021, 2023 and 2024 - five papers. Nothing else in geography comes close. The National Highways Development Project was built in phases. Phase I was the Golden Quadrilateral joining the four metros. Phase II was the two long corridors that cut the country north-south and east-west. Where those two corridors cross is the single most asked fact in this subject.

- India's road network is the second largest in the world.
- National highways are about 1.46 lakh km (2024-25).
- They are roughly 2 per cent of total road length but carry about 40 per cent of road traffic.
- NHAI was constituted under an Act of 1988 and became operational in 1995.
- Bharatmala Pariyojana (2017) Phase-I covers 34,800 km.

The Golden Quadrilateral circuit, with its highway numbers

Delhi to Kolkata - NH-19 → Kolkata to Chennai - NH-16 →

Chennai to Mumbai - NH-48 → Mumbai to Delhi - NH-48 →

Total 5,846 km, NHDP Phase I, completed 2012

ASKED IN RAS

The Golden Quadrilateral has been asked three ways. RAS Pre 2013 - which four cities does it connect? Delhi, Kolkata, Chennai and Mumbai. RAS Pre 2023 - its length? 5,846 km. RAS Pre 2016 - a match-the-following on segment to highway number: Delhi-Kolkata NH-19, Kolkata-Chennai NH-16, Chennai-Mumbai NH-48, Srinagar-Kanniyakumari NH-44.

The two corridors at a glance

Corridor	Route	Mainly which NH
North-South Corridor	Srinagar to Kanniyakumari, with a Salem-Kochi spur	NH-44

Corridor	Route	Mainly which NH
East-West Corridor	Silchar to Porbandar	NH-27
Where they cross	Jhansi, Uttar Pradesh	Both - NHDP Phase II

How to picture the crossing

North-South starts at Srinagar and runs down through Delhi, Agra, Gwalior →

East-West starts at Silchar and runs west through Lucknow, Kanpur →

The two meet at JHANSI in Uttar Pradesh →

North-South goes on to Nagpur, Bengaluru, Kanniyakumari →

East-West goes on to Ahmedabad and ends at Porbandar

North-South vs East-West Corridor

North-South Corridor	East-West Corridor
Srinagar (Jammu and Kashmir) to Kanniyakumari (Tamil Nadu)	Silchar (Assam) to Porbandar (Gujarat)
Largely NH-44	Largely NH-27
Has a spur from Salem to Kochi	No spur
NH-44 is India's longest highway, 4,113 km	NH-27 is the second longest, about 3,507 km
Runs top to bottom of the country	Runs from the north-eastern hills to the Arabian Sea

ASKED IN RAS

Asked in RAS Pre 2016 and again in RAS Pre 2021 - where do the North-South and East-West corridors intersect? Answer: Jhansi. In 2021 it came as a three-statement question in which all three statements were correct. If you learn one word from this chapter, learn Jhansi.

ASKED IN RAS

Asked in RAS Pre 2024 - the East-West Corridor connects which two places? Answer: Silchar and Porbandar. Mind the direction of the pair: Silchar is in Assam in the east, Porbandar in Gujarat in the west.

- NH-44 (4,113 km) is the longest national highway in India; NH-27 (about 3,507 km) is second.
- NH-66 runs along the west coast.
- Under the 2010 renumbering, north-south highways get EVEN numbers increasing from east to west.
- East-west highways get ODD numbers increasing from north to south.

YAAD RAKHO

Numbering rule in four words: even goes down, odd goes across. Even numbers are north-south routes, counted westwards from the east coast; odd numbers are east-west routes, counted southwards from the north.

TRAP

Old numbers still float in the options. NH-2 is now NH-19 (Delhi-Kolkata), NH-5 is NH-16 (Kolkata-Chennai), NH-4 and NH-8 are now NH-48, NH-7 is now NH-44. So an option reading 'NH-48 - Delhi to Kolkata' is wrong: NH-48 is the Delhi-Mumbai-Chennai side.

Railways, ports, waterways and airports

The transport tail of the chapter is pure one-liners. Firsts, counts and superlatives - that is all that gets asked. Learn the count of railway zones, the count of major ports and the first three national waterways, and you have covered it.

- India's first passenger train ran from Bori Bunder to Thane, 34 km, on 16 April 1853.
- Indian Railways has 19 zones; Kolkata Metro is counted as one of them.
- The newest zone is South Coast Railway, headquartered at Visakhapatnam.
- India has 12 major ports, administered by the Centre.
- Mundra is a large port but not a 'major port' in the legal sense - it is the standard wrong option.
- Visakhapatnam handles the Bailadila iron ore exports to Japan.

CURRENT LINK

Current link, as of September 2026: the South Coast Railway zone became functional on 1 June 2026 with headquarters at Visakhapatnam and four divisions - Guntakal, Guntur, Vijayawada and Visakhapatnam. It takes Indian Railways to 19 zones. Any 2026 question about 'the newest railway zone' points here.

Port - state - identifier

Port	State	Identifier
Jawaharlal Nehru (Nhava Sheva)	Maharashtra	Largest container-handling major port
Syama Prasad Mookerjee (Kolkata)	West Bengal	Only major riverine port
Kamarajar (Ennore)	Tamil Nadu	First corporatised major port
Deendayal	Gujarat	The renamed Kandla port

Port	State	Identifier
V. O. Chidambaranar	Tamil Nadu	The renamed Tuticorin port
Paradip	Odisha	Bulk cargo, iron ore
Mormugao	Goa	Iron ore exports
Vadhavan (Palghar)	Maharashtra	Approved in 2024 as a new major port

The first three national waterways

Waterway	River / canal	Stretch
NW-1	Ganga	Haldia to Prayagraj, 1,620 km
NW-2	Brahmaputra	Dhubri to Sadiya, 891 km
NW-3	West Coast Canal	Kollam to Kottapuram, 205 km

- The National Waterways Act, 2016 declared 111 national waterways.
- The Inland Waterways Authority of India was set up in 1986; its headquarters is at Noida, not Delhi.
- NW-1 ends at Prayagraj, not Varanasi - both are single-word substitutions RPSC likes.
- Indira Gandhi International, Delhi, is India's busiest airport.

REVISION RECAP

1. Iron ore - Odisha is the largest producer; belts are Odisha-Jharkhand, Durg-Bastar-Chandrapur (Bailadila), Bellary-Chitradurga (Kudremukh) and Maharashtra-Goa.
2. Manganese - MP leads production (about 31%) but Odisha holds the largest reserves (about 34%); copper works the same way, MP produces and Rajasthan holds the reserves.
3. Bauxite Odisha (Panchpatmali, Koraput); lead-zinc almost entirely Rajasthan (Rampura Agucha, Zawar, Rajpura-Dariba) under Hindustan Zinc; chromite Sukinda; gold Hutti and Kolar.
4. Rock phosphate Jhamarkotra, gypsum Nagaur-Bikaner, limestone Rajasthan largest producer with Karnataka's largest reserves, diamond Panna, uranium Jaduguda and Tummalapalle.

5. Coal - 96 per cent Gondwana; Jharia largest and coking, Raniganj oldest (1774); Jharkhand largest reserves, Odisha leads production. Rajasthan has lignite, not coal.
6. Lignite - Neyveli largest; Kapurdi-Jalipa and Barsingsar in Rajasthan; Panandhro in Gujarat.
7. Mumbai High (1974, Sagar Samrat) largest oil field; Mangala in the Barmer basin; Digboi (1901) oldest refinery; Jamnagar the world's largest complex; Pachpadra refinery at Balotra.
8. Nuclear - Tarapur oldest (1969), Rawatbhata Canadian-aided, Kudankulam largest and Russian-aided, Kakrapar the first indigenous 700 MWe PHWR.
9. Steel - Bhilai and Bokaro USSR, Rourkela West Germany, Durgapur United Kingdom; Jamshedpur first and private; Visakhapatnam the only shore-based plant.
10. Nicknames - Ahmedabad Manchester of India, Coimbatore Manchester of the South, Chennai Detroit of India, Bengaluru Silicon Valley of India.
11. Eight industrial regions; Hugli is the oldest and jute-based, Chotanagpur is iron and steel, Kollam-Thiruvananthapuram is plantation-based.
12. DFCs - Western Dadri to JNPT (Japanese aid), Eastern Sahnewal to Sonnagar (World Bank); DMIC follows the Western DFC and about 39 per cent of it lies in Rajasthan.
13. Golden Quadrilateral - 5,846 km, NHDP Phase I, completed 2012; Delhi-Kolkata NH-19, Kolkata-Chennai NH-16, Chennai-Mumbai and Mumbai-Delhi NH-48.
14. North-South Corridor Srinagar to Kanniyakumari (NH-44, Salem-Kochi spur); East-West Corridor Silchar to Porbandar (NH-27); the two cross at JHANSI - asked in five papers.
15. NH-44 longest at 4,113 km, NH-27 second at about 3,507 km, NH-66 along the west coast; even numbers run north-south increasing east to west, odd numbers east-west increasing north to south.
16. National highways about 1.46 lakh km, 2 per cent of road length carrying 40 per cent of traffic; NHAI Act 1988, operational 1995; Bharatmala Phase-I 34,800 km.
17. Railways - first train Bori Bunder to Thane, 16 April 1853; 19 zones including Kolkata Metro; South Coast Railway functional 1 June 2026 at Visakhapatnam.
18. 12 major ports - JNPT largest container, Kolkata the only riverine, Kamarajar first corporatised, Deendayal is Kandla, Vadhavan approved 2024; NW-1 Ganga, NW-2 Brahmaputra, NW-3 West Coast Canal; IWAI at Noida.